INCH-POUND

MIL-C-70569B (AR)
27 September 1993
SUPERSEDING
(See 6.6)

MILITARY SPECIFICATION

CIRCUIT BOARD, SYNCHRO INTERFACE (PIVADS): 9360871

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the fire control processor synchro interface circuit card assembly referred to as the Synchro Interface Board (SIB). The SIB is used to convert gun and sight position signals (azimuth and elevation) from analog to digital format and antenna position command signals from digital to analog format. The SIB is a 6 by 11 1/2 inch double-layered, multiwire circuit card mounted vertically in the Fire Control Processor (FCP) and hard wired to the Servo Controller Board (SCB). The SIB contains 24 microcircuits and resistive and capacitor circuitry (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1285

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SPECIFICATIONS

MILITARY

MIL-F-13926 - Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-I-45607 - Inspection Equipment, Acquisition, Maintenance and Disposition of

STANDARDS

MILITARY

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-810 - Environmental Test Methods and Engineering Guidelines

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360871 - Synchro Interface Assembly
9360873 - Synchro Interface Schematic
TPS 9361336- Test Program Set Manual
9397371 - Interface Device B (IDB)
9397497 - Cable Assembly W1
9397523 - Test Plug P6
9397524 - Test Plug P7
9397525 - Test Plug P8
9397526 - Test Plug P9
12950877 - Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 9360871 - Packaging of Synchro Interface Assembly

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(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

ANSI/IPC-DW-425 - Design and End Product Requirements for Discrete Wiring Boards

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360871, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the SIB shall meet the requirements of MIL-F-13926 and ANSI/IPC-DW-425 as follows:

3.2.1 MIL-F-13926.

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

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c. Part identification and marking.

d. Workmanship.

3.2.2 ANSI/IPC-DW-425.

a. IPC-DW-425/11 Type C1 plated through connection.

3.3 Environmental service conditions.

3.3.1 Vibration. The SIB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The SIB shall not suffer damage and shall meet the requirements of 3.4 and 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10 percent, 11 milliseconds ± 10 percent, half-sine wave applied in each direction of 3 mutually perpendicular axes.

3.3.3 Altitude operating. The SIB shall meet the requirements of 3.4 and 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The SIB shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +155°F.

3.3.4.2 Non-operating (storage). The SIB shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours and +160°F for a period of 4 hours (after temperature stabilization), and shall meet the requirements of 3.4 and 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The SIB shall meet the requirements of 3.4 and 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.4 Electrical. Each conductor shall be electrically continuous as specified on the schematic diagram 9360873.

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3.5 Performance.

3.5.1 Functional description. The primary functions of the SIB shall be digital to synchro (D/S) conversion of the antenna position signal, synchro to digital (S/D) conversion of the sight and cannon position signals, end-around testing (EAT), and receiving position information from the sight and the cannon. The SIB shall have the capability of testing the D/S's and S/D's at a system BIT level. The SIB shall also move digital data to and from a test buffer. All control logic, data conversion timing and initiation on the SIB shall be controlled by signals from the SCB. The SIB shall utilize 9 control signals from the SCB, 8 inputs and 1 output. The SIB shall communicate with the other circuit card assemblies in the FCP through the SCB. See Figure 1 for the functional block diagram.

3.5.2 Power requirements. SIB power requirements shall be as listed in Table I and available on pins as specified in 3.5.7.1.

TABLE I. SIB power requirements.

	VOLTAGE (Vdc) MIN	TYP	MAX	AMPS Adc	IMPD OHMS	RIPPLE V p-p
+ 5 V	4.75	5.00	5.25	1.5	<.033	<.125
+15 V	14.25	15.00	15.75	.533	<.033	<.375
-15 V	-14.25	-15.00	-15.75	.533	<.033	<.375
+24 V	19.20	24.00	28.80	.6	---	---

3.5.3 Clock requirements. The SIB assembly clock (800 kHz) signal applied at P1-28 shall have TTL voltage levels per 3.5.4. The clock frequency shall be 800 kHz + 120 Hz.

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3.5.4 TTL voltage levels. The TTL voltage levels shall be as listed in Table II:

TABLE II. TTL voltage levels.

TTL SIGNAL	MIN	TYP	MAX	UNITS
Voltage Output Low (VOL)	---	0.2	0.4	V
Voltage Output High (VOH)	2.4	3.4	---	V
Voltage Input Low (VIL)	---	---	0.7	V
Voltage Input High (VIH)	2.0	---	---	V

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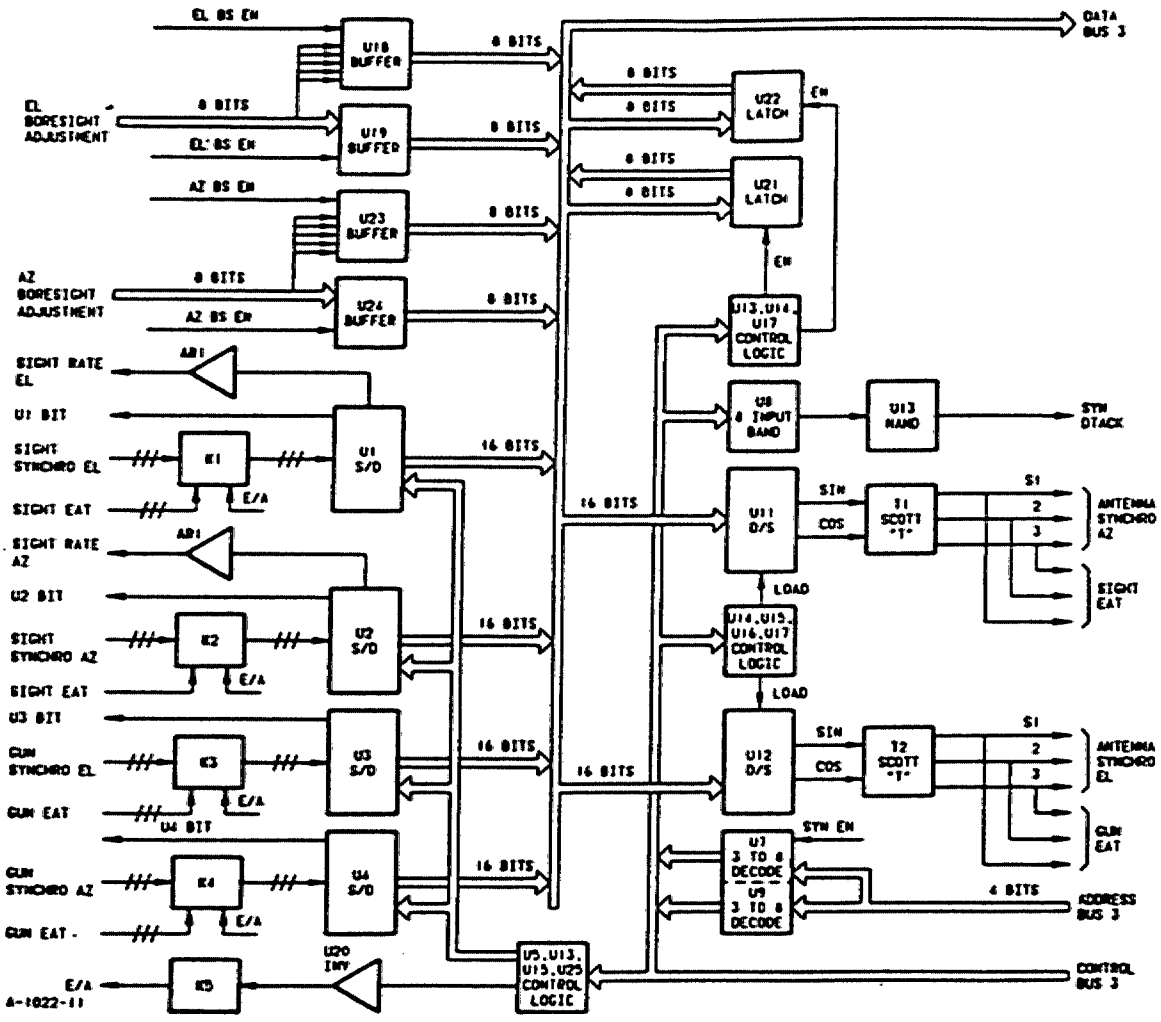


FIGURE 1. SIB functional block diagram.

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3.5.5 Synchro signal. Each synchro signal shall be a 3-phase 400 Hz signal. The electrical angle "0" associated with each synchro signal shall represent the angle, displaced in a positive direction from the synchro zero, which satisfies the relative magnitude and polarities of the secondary voltages of an ideal synchro transmitter or receiver in accordance with the following equations:

$$E(S1S3) = E(R2R1) \sin \varnothing$$
$$E(S3S2) = E(R2R1) \sin (\varnothing + 120)$$
$$E(S2S1) = E(R2R1) \sin (\varnothing + 240)$$

Where:

The terminal sequence as indicated in the parentheses designates the sense of the voltage vector.

E(R2R1) is the voltage between lines R2 and R1.
E(S1S3) is the voltage between lines S1 and S3.
E(S3S2) is the voltage between lines S3 and S2.
E(S2S1) is the voltage between lines S2 and S1.

3.5.6 SIB functions. Control logic, data conversion timing and initiation of the SIB are controlled by nine signals passed between the SIB and the SCB. The eight input signals (SYN EN (P1-29), B3 A1 (P1-66), B3 A2 (P1-65), B3 A3 (P1-26), B3 A4 (P1-25), LDS (P1-70), R/W (P1-30), CLOCK (P1-28)) are decoded by the SIB into the necessary timing signals. The one output signal (SYN DTACK (P1-69)) informs the SCB that the signal sent to the SIB has been received. SIB functions are determined by signals A1, A2, A3, and A4 as specified in Table III, and commence with the receipt of SYN EN.

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TABLE III. SIB function addresses.

B3 A4	B3 A3	B3 A2	B3 A1	FUNCTION	CYCLE
0	0	0	0	ANTENNA AZ D/S	3.5.6.1
0	0	0	1	ANTENNA EL D/S	3.5.6.1
0	0	1	0	NOT USED	
0	0	1	1	NOT USED	
0	1	0	0	NOT USED	
0	1	0	1	NOT USED	
0	1	1	0	NOT USED	
0	1	1	1	$\overline{\text{BITE ON}}$	3.5.6.3
1	0	0	0	$\overline{\text{BITE OFF}}$	3.5.6.4
1	0	0	1	GUN AZ S/D	3.5.6.2
1	0	1	0	GUN EL S/D	3.5.6.2
1	0	1	1	SIGHT AZ S/D	3.5.6.2
1	1	0	0	SIGHT EL S/D	3.5.6.2
1	1	0	1	AZ BORESIGHT READ	
1	1	1	0	EL BORESIGHT READ	
1	1	1	1	NOT USED	

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3.5.6.1 Digital to Synchro conversion (D/S) cycle. D/S timing shall be as shown in Figure 2. The D/S cycle requires that the 800 KHz and appropriate address signals be supplied to the SIB at time of receipt of the high to low transition of SYN EN. The address and lock signals shall be decoded by the SIB to produce the following signals:

Signal	Test Point	n = 1	n = 2
$\overline{\text{D/Sn EN}}$		J1-20	J1-19
$\overline{\text{D/S EN}}$	J1-29		
D/Sn GATED EN		J1-26	J1-25
D/S WRITE EN	J1-27		
D/Sn LD		J1-34	J1-35
$\overline{\text{ACK}}$	J1-33		
SYN DTACK	J1-17		
$\overline{\text{SYN DTACK}}$	P1-69		

Where: n = 1 or 2 depending on the address received.

The D/S is accomplished by a digital-to-resolver converter (DRC) in conjunction with a Scott-T transformer. When the DRC receives a high on D/Sn LD, the DRC shall convert the 16-bit data word (TTL per 3.5.4) on lines D15-0 into a resolver format output satisfying the following conditions:

$$E(U1n\text{-}5,28) = E(R1R2)\sin \varnothing$$
$$E(U1n\text{-}4,28) = E(R1R2)\cos \varnothing$$

Where: n= 1 or 2

E(U1n-5,28) is the voltage between microcircuit U1n pins 5 and 28
E(U1n-4,28) is the voltage between microcircuit U1n pins 4 and 28
E(R1R2) is the voltage between P2-28 and P2-29
$\varnothing$ is the angle represented by the 16-bit data word as specified in Table IV.

The resolver format signal shall then be converted to a 90 ± 9 Vrms L-L 400 ± 0.1 percent Hz synchro format signal per 3.5.5 by the Scott-T transformer Tn and output on P2-33, 34, 35 (30, 31, 32). Overall digital input to synchro output accuracy shall be ± 0.16667 degrees within a settling time of 40 µs maximum.

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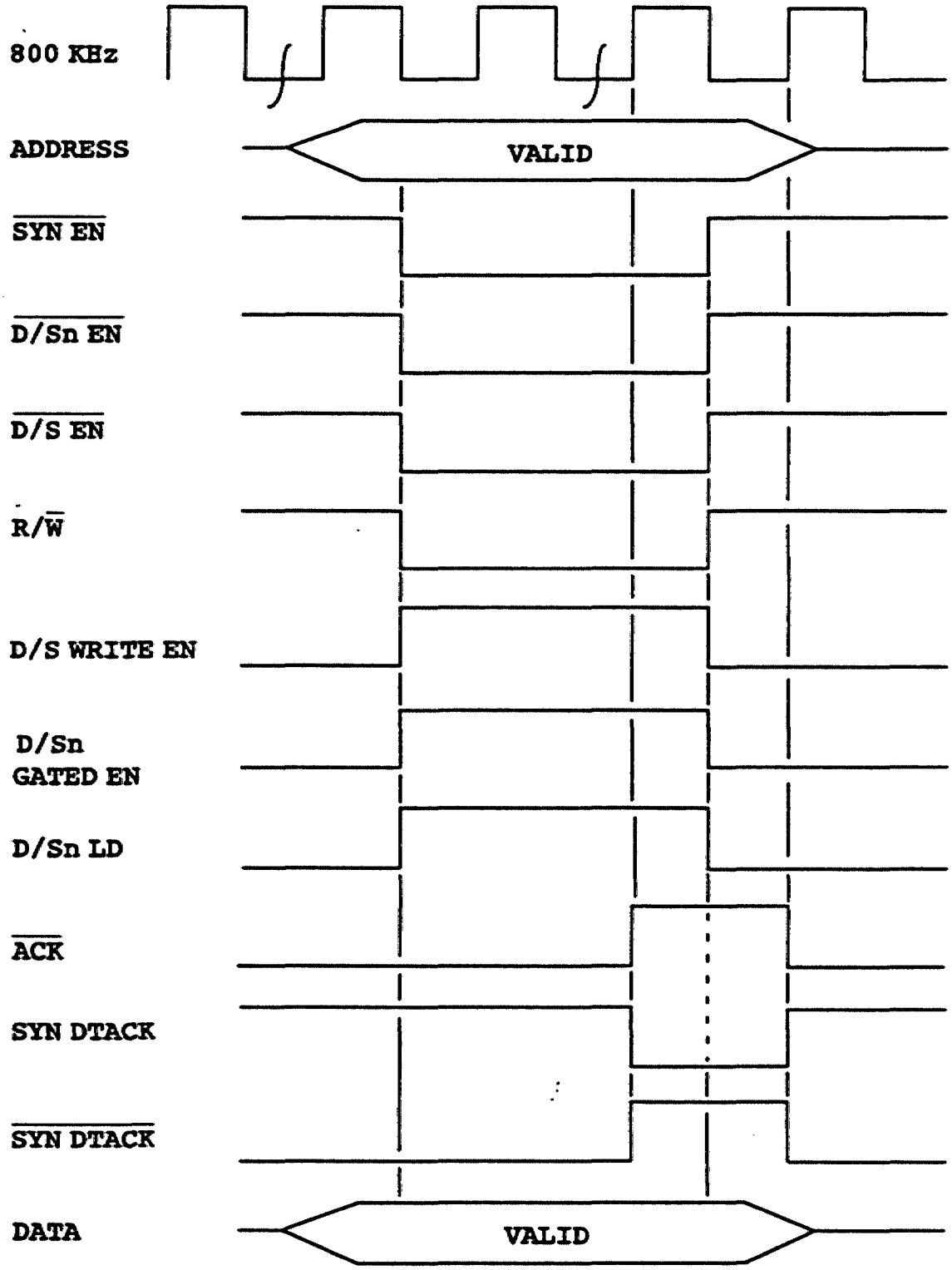


FIGURE 2. D/S cycle timing.

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TABLE IV. **Binary representation of angles.**

Bit No.		Degrees
MSB	1	180
	2	90
	3	45
	4	22.5
	5	11.25
	6	5.625
	7	2.8125
	8	1.40625
	9	0.70312
	10	0.35156
	11	0.17578
	12	0.08789
	13	0.04395
	14	0.02197
	15	0.01099
LSB	16	0.00549

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3.5.6.2 Synchro to Digital conversion (S/D) cycle. S/D timing shall be as shown in Figure 3. The S/D cycle requires that the 800 KHz and appropriate address signals be supplied to the SIB at time of receipt of the high to low transition of SYN EN. The address and clock signals shall be decoded by the SIB to produce the following signals:

Signal	Test Point	n = 1	n = 2	n = 3	n = 4
$\overline{\text{S/Dn EN}}$		J1-12	J1-13	J1-14	J1-15
S/Dn BUSY		J1-1	J1-2	J1-3	J1-4
$\overline{\text{S/Dn DR}}$		J1-5	J1-6	J1-7	J1-8
$\overline{\text{S/D DTACK}}$	J1-44				
SYN DTACK	J1-17				
$\overline{\text{SYN DTACK}}$	P1-69				

where n = 1, 2, 3, 4 depending on the address received.

The S/D circuit shall be capable of detecting the electrical angle of the 90 ± 30 percent Vrms L-L, 400 ± 0.1 percent Hz, 10 percent maximum harmonic distortion synchro input signal and the 115 ± 5 percent Vrms, 400 ± 0.1 percent Hz sine wave reference signal with 10 percent maximum harmonic distortion found on P2-26, 27, and converting the synchro electrical angle into a 16-bit data word (TTL per 3.5.4) with an accuracy of ± 0.02167 degrees for azimuth and elevation sight synchro input angles and ± 0.08667 degrees for azimuth and elevation gun synchro input angles. The 16-bit word shall represent the electrical angle by binary weighting as specified in Table IV. For a 179-degree step input change, synchro to digital settling time shall be 264 milliseconds maximum to 1 LSB accuracy. For a step input change of Ø/-1.4 degrees, synchro to digital settling time shall be 35 milliseconds maximum to 1 LSB accuracy. Within 3 µsec of receiving a low on the S/Dn EN line, the S/D microcircuit shall 'freeze' the digital output on data bus D15-0 and pull S/Dn BUSY low for 1.5 µsec, signifying the data on the bus is valid.

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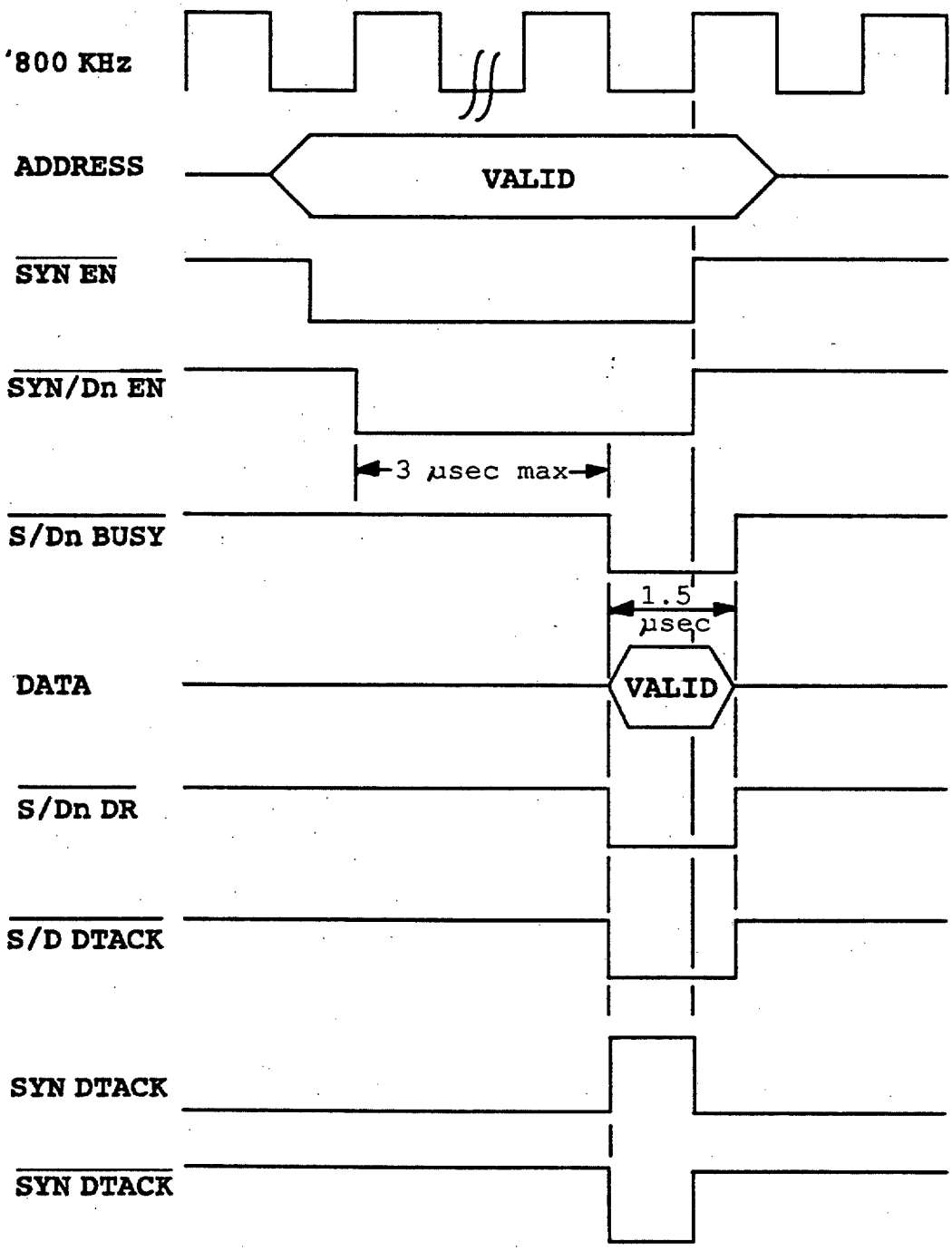


FIGURE 3. S/D cycle timing.

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3.5.6.3 BITE ON cycle. BITE ON cycle timing shall be as shown in Figure 4. The SIB requires the appropriate address signals be supplied at the time that SIB receives the high to low transition of SYN EN. The address and clock signals shall be decoded by the SIB to produce the following signals:

Signal	Test Point
BITE ON	J1-18
E/A1	J1-41
BITE ON	J1-38
DATA LATCH READ	J1-23
SYN DTACK	J1-17
SYN DTACK	P1-69

The E/A1 signal, when latched high, shall energize relay K5, which in turn shall energize relays K1-4, making the outputs of the D/S circuits the inputs to the S/D circuit as specified in Table V. The DATA LATCH READ signal shall enable the output of data buffers U21 and U22, allowing the data stored in the buffer to be read on data bus D15-0.

TABLE V. S/D inputs.

	S/D INPUT	
S/D	**E/A1 HIGH**	**E/A1 LO (D/S outputs)**
U1	SIGHT SYNCHRO EL (P2-17, 18, 19)	ANTENNA SYNCHRO AZ (P2-33, 34, 35)
U2	SIGHT SYNCHRO AZ (P2-14, 15, 16)	ANTENNA SYNCHRO AZ (P2-33, 34, 35)
U3	GUN SYNCHRO EL (P2-23, 24, 25)	ANTENNA SYNCHRO EL (P2-30, 31, 32)
U4	GUN SYNCHRO EL (P2-20, 21, 22)	ANTENNA SYNCHRO EL (P2-30, 31, 32)

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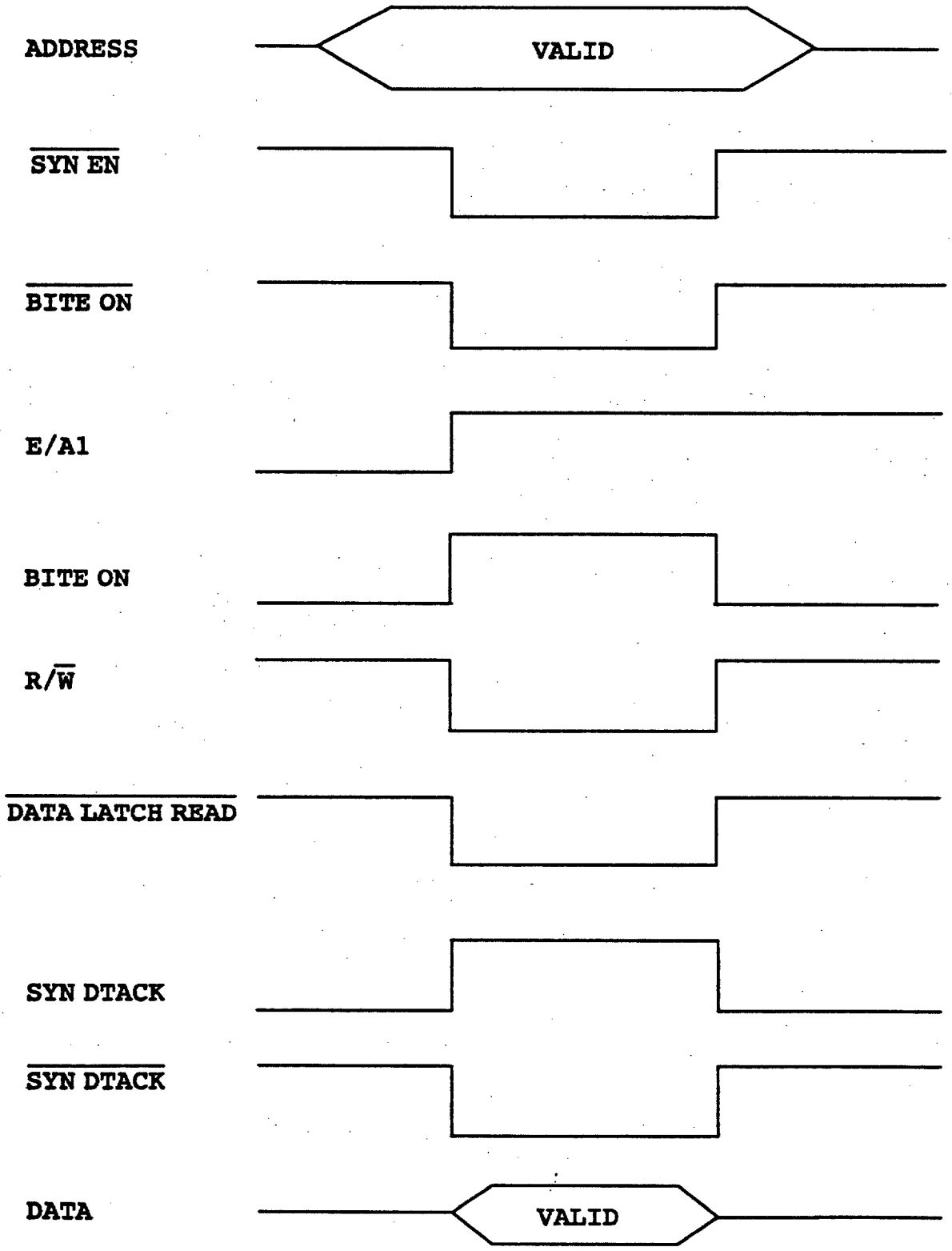


FIGURE 4. BITE ON cycle timing.

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3.5.6.4 BITE OFF cycle. BITE OFF cycle timing shall be as shown in Figure 5. The SIB requires the appropriate address signals be supplied at the time the SIB receives the high to low transition of SYN EN. The address and clock signals shall be decoded by the SIB to produce the following signals:

Signal	Test Point
BITE OFF	J1-16
EAT OFF	J1-31
E/A 1	J1-41
DATA LATCH WRITE EN	J1-37
DATA LATCH WRITE	J1-28
SYN DTACK	J1-17
SYN DTACK	P1-69

The E/A1 signal, when latched low, shall de-energize relay K5, which in turn, shall de-energize relays K1-4, switching the inputs to the S/D circuits as specified in Table V. The DATA LATCH WRITE signal shall write the data on data bus D15-0 to buffers U21 and U22.

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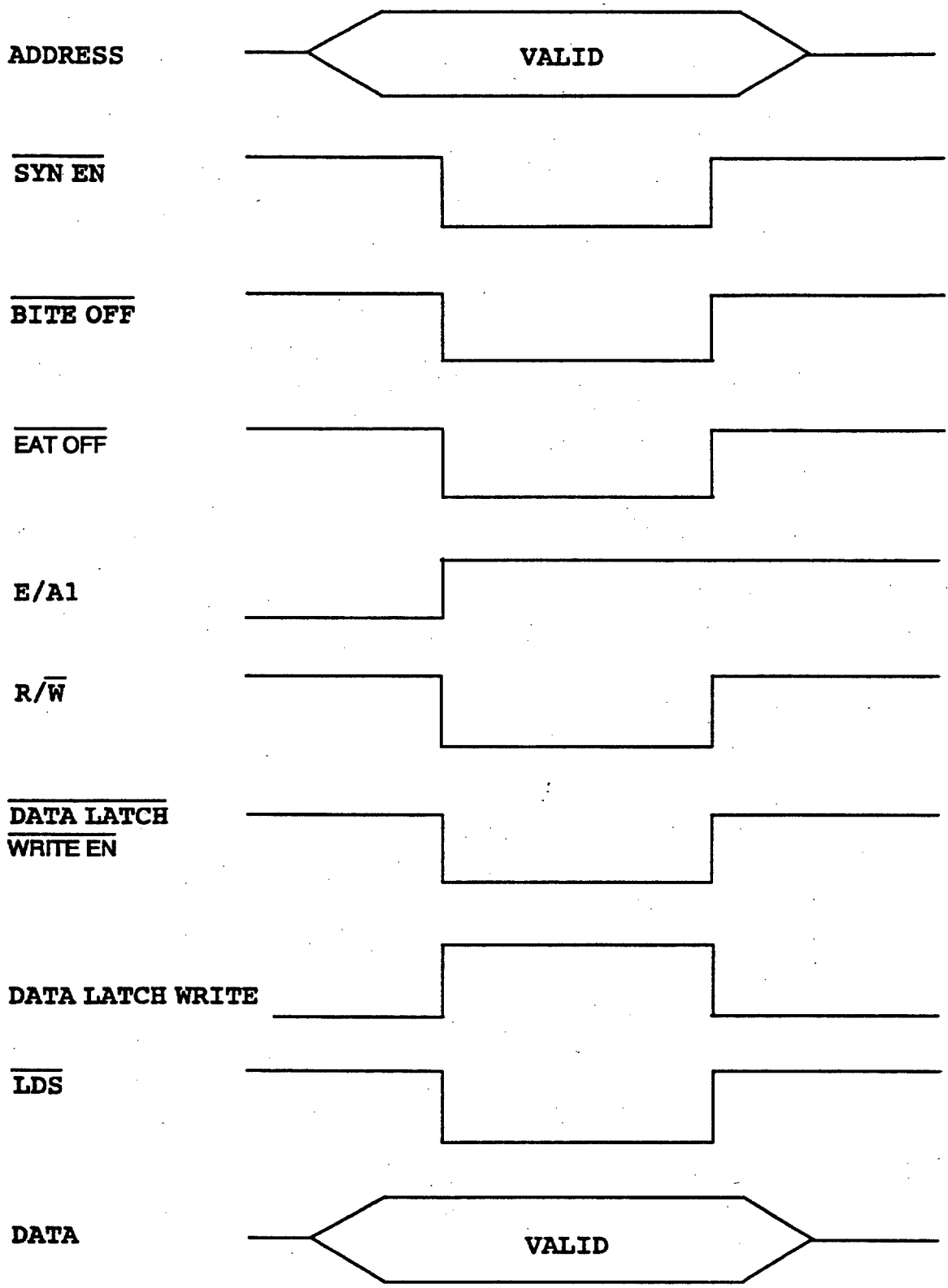


FIGURE 5. BITE OFF cycle timing.

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3.5.7 Input/output requirements.

3.5.7.1 SIB power supply pins. The SIB power supply pins shall be as listed in Table VI.

3.5.7.2 SIB connector P1 inputs. The SIB connector P1 inputs shall be as listed in Table VII and TTL levels per 3.5.4.

3.5.7.3 SIB connector P1 inputs/outputs. The SIB connector P1 inputs/outputs shall be as listed in Table VIII with TTL levels per 3.5.4.

3.5.7.4 SIB connector P2 inputs. The SIB connector P2 inputs shall be as listed in Table IX.

3.5.7.5 SIB connector P2 outputs. The SIB connector P2 outputs shall be 0-90 Vrms signals on the pins listed in Table X.

3.5.7.6 SIB test points. The SIB test points shall be as listed in Table XI with TTL levels per 3.5.4.

3.5.7.7 SIB spare pins. The SIB spare pins shall be specified in Table XII.

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TABLE VI. Power supply, return, and ground pins.

CONNECTOR PIN	SIGNAL NAME
P1-9	+5 V RTN
P1-18	+5 V RTN
P1-32	+5 V
P1-33	+5 V
P1-34	+5 V
P1-35	+5 V RTN
P1-36	+5 V RTN
P1-37	+5 V RTN
P1-38	± 15 V RTN
P1-39	-15 V
P1-40	+15 V
P1-49	+5 V RTN
P1-58	+5 V RTN
P1-72	+5 V
P1-73	+5 V
P1-74	+5 V
P1-75	+5 V RTN
P1-76	+5 V RTN
P1-77	+5 V RTN
P1-78	±15 V RTN
P1-79	-15 V
P1-80	+15 V
P2-3	+24 V
P2-4	GND
P2-5	+5 V
P2-38	+24 V RTN
P2-39	GND
P2-40	+5 V

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TABLE VII. Connector P1 inputs.

PIN NO.	SIGNAL NAME
25	B3 - A4
26	B3 - A3
28	CLOCK (800 KHz)
29	$\overline{\text{SYN EN}}$
30	R/$\overline{\text{W}}$
65	B3 - A2
66	B3 - A1
70	$\overline{\text{LDS}}$

TABLE VIII. Connector P1 inputs/outputs.

PIN NO.	SIGNAL NAME
10	D15
50	D14
11	D13
51	D12
12	D11
52	D10
13	D9
53	D8
14	D7
54	D6
15	D5
55	D4
16	D3
56	D2
17	D1
57	D0
69	$\overline{\text{SYN DTACK}}$

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TABLE IX. Connector P2 inputs.

PIN NO.	SIGNAL NAME	SIGNAL
6	B7 EL BORESIGHT ADJUST	NOT USED
7	B6 EL BORESIGHT ADJUST	NOT USED
8	B5 EL BORESIGHT ADJUST	NOT USED
9	B4 EL BORESIGHT ADJUST	NOT USED
10	B3 EL BORESIGHT ADJUST	NOT USED
11	B2 EL BORESIGHT ADJUST	NOT USED
12	B1 EL BORESIGHT ADJUST	NOT USED
14	S1 SIGHT SYNCHRO AZ	0-90 VRMS 400 Hz
15	S2 SIGHT SYNCHRO AZ	0-90 VRMS 400 Hz
16	S3 SIGHT SYNCHRO AZ	0-90 VRMS 400 Hz
17	S1 SIGHT SYNCHRO EL	0-90 VRMS 400 Hz
18	S2 SIGHT SYNCHRO EL	0-90 VRMS 400 Hz
19	S3 SIGHT SYNCHRO EL	0-90 VRMS 400 Hz
20	S1 GUN SYNCHRO AZ	0-90 VRMS 400 Hz
21	S2 GUN SYNCHRO AZ	0-90 VRMS 400 Hz
22	S3 GUN SYNCHRO AZ	0-90 VRMS 400 Hz
23	S1 GUN SYNCHRO EL	0-90 VRMS 400 Hz
24	S2 GUN SYNCHRO EL	0-90 VRMS 400 Hz
25	S3 GUN SYNCHRO EL	0-90 VRMS 400Hz
26	R1 SD REFERENCE	115 VRMS 400 Hz
27	R2 SD REFERENCE	115 VRMS 400 Hz
28	R1 SD REFERENCE	115 VRMS 400 Hz
29	R2 SD REFERENCE	115 VRMS 400 Hz
41	B7 AZ BORESIGHT ADJUST	NOT USED
42	B6 AZ BORESIGHT ADJUST	NOT USED
43	B5 AZ BORESIGHT ADJUST	NOT USED
44	B4 AZ BORESIGHT ADJUST	NOT USED
45	B3 AZ BORESIGHT ADJUST	NOT USED
46	B2 AZ BORESIGHT ADJUST	NOT USED
47	B1 AZ BORESIGHT ADJUST	NOT USED
53	S/D1 - BIT	TTL per 3.5.4
54	S/D2 - BIT	TTL per 3.5.4
55	S/D3 - BIT	TTL per 3.5.4
56	S/D4 - BIT	TTL per 3.5.4

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TABLE X. Connector P2 outputs.

PIN NO.	SIGNAL NAME
30	S1 ANTENNA SNYCHRO EL
31	S2 ANTENNA SYNCHRO EL
32	S3 ANTENNA SYNCHRO EL
33	S1 ANTENNA SYNCHRO AZ
34	S2 ANTENNA SYNCHRO AZ
35	S3 ANTENNA SYNCHRO AZ

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TABLE XI. Connector J1 test points.

PIN	SIGNAL NAME	PIN	SIGNAL NAME	PIN	SIGNAL NAME
1	S/D1 BUSY	16	$\overline{\text{BITE OFF}}$	31	$\overline{\text{EAT OFF}}$
2	S/D2 BUSY	17	SYN DTACK	32	Q
3	S/D3 BUSY	18	$\overline{\text{BITE ON}}$	33	$\overline{\text{ACK}}$
4	S/D4 BUSY	19	$\overline{\text{D/S2 EN}}$	34	D/S1 LD
5	$\overline{\text{S/D1 DR}}$	20	$\overline{\text{D/S1 EN}}$	35	D/S2 LD
6	$\overline{\text{S/D2 DR}}$	21	Q1	36	$\overline{\text{DATA LATCH}}$
7	$\overline{\text{S/D3 DR}}$	22	Q2	37	$\overline{\text{WRITE EN}}$
8	$\overline{\text{S/D4 DR}}$	23	$\overline{\text{DATA LATCH READ}}$	38	BITE ON
9	U9 EN	24	$\overline{\text{U26 SET}}$	39	$\overline{\text{800 KHz}}$
10	$\overline{\text{EL BS EN}}$	25	D/S2 GATED EN	40	RESET
11	$\overline{\text{AZ BS EN}}$	26	D/S1 GATED EN	41	E/A1
12	$\overline{\text{S/D1 EN}}$	27	D/S WRITE EN	42	$\overline{\text{U16 SET}}$
13	$\overline{\text{S/D2 EN}}$	28	DATA LATCH WRITE	43	S/D DR
14	$\overline{\text{S/D3 EN}}$	29	$\overline{\text{D/S EN}}$	44	$\overline{\text{S/D DTACK}}$
15	$\overline{\text{S/D4 EN}}$			45	$\overline{\text{RESET}}$

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TABLE XII. Spare connector pins.

CONNECTOR PIN
J1-30
J1-36
J1-46
J1-47
J1-48
J1-49
J1-50
P1-1
P1-2
P1-3
P1-4
P1-5
P1-6
P1-7
P1-19
P1-20
P1-21
P1-22
P1-23
P1-24
P1-27
P1-31
P1-41
P1-42
P1-43
P1-44
P1-45
P1-46
P1-59

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TABLE XII. Spare connector pins. (continued)

CONNECTOR PIN
P1-60
P1-61
P1-62
P1-63
P1-64
P1-67
P1-68
P1-71
P2-1
P2-2
P2-13
P2-36
P2-37
P2-48
P2-49
P2-50
P2-51
P2-52
P2-57
P2-58
P2-59
P2-60
P2-61
P2-62
P2-63
P2-64
P2-65
P2-66
P2-67
P2-68
P2-69
P2-70

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3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

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4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	90 percent RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
Unit SIB	9360871	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in Table XIII (Functional tests) and Table XIV (Special sampling) herein and to any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one

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manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 herein shall be performed as specified in 4.4.2.3 and 4.5.

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4.4.2.3 Functional tests. The tests in Table XIII shall be performed on a 100 percent basis.

TABLE XIII. Functional tests (100 percent inspection).

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 101. Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

4.5 Special sampling. One SIB out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table XIII and shall then meet the requirements and tests in Table XIV. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

TABLE XIV. Special sampling.

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 103. Vibration 104. Shock 105. Altitude 106. Temperature 107. Humidity	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

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4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

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4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table XV (Test equipment list) shall be utilized in the performance of the applicable test methods and procedures as specified in 4.7. The following items are required to test the SIB and shall be assembled at the AN/USM-465A Test Station prior to test execution. Prior operation of interface devices and test equipment shall be verified before using same to run diagnostic tests.

TABLE XV. Test equipment list.

Item Number	Description
1. Digital Card Tester	AN/USM-465A
2. ComCode Program Tape	Version 2.0
3. Test Program Set Manual	TPS 9361336 (GFE)
4. Test Program Set Tape	TPT 9361336 (GFE)
5. Interface Device B (IDB)	9397371 with Cable Assembly W1 (9397497) and test plugs P6 (9397523), P7 (9397524), P8 (9397525), and P9 (9397526) (GFE)
6. Oscilloscope	Input impedance of 1 Mohm or greater, single sweep capability, with probe .0 to ± 100 Vdc and 10 MHz capability minimum
7. Synchro angle indicator	Minimum accuracy of ± 0.3 ARC minutes (0.0050 degrees). The meter must be set for bipolar operation.

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4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein. In Step 1 of Procedure I, only steps 4 through 8 of paragraph II-2.1 apply. The SIB shall be attached to the vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the SIB shall be tested as specified in 4.7.2 and 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10 percent amplitude and 11 ms ± 10 percent duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10 percent.

d. The shock measuring system shall have a flat frequency response within ± 10 percent from 20 to 2,000 hertz.

After the application of all the shock impulses, the SIB shall be tested as specified in 4.7.2 and 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude (operating). This test shall be performed as specified in Procedure II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.3.

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4.7.1.4 Temperature.

4.7.1.4.1 High (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2, 4.7.3, for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.4.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.2 and 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the synchro interface board shall be tested using the test equipment listed in Table XV in accordance with the SIB Test Program Set, drawing TPS 9361336 to verify the requirements of Section 3.5.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of drawing 12950877 as specified in paragraph 3.7.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360871. The level of protection shall be as specified in the procurement documents.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The synchro interface circuit card assembly shall be used in the fire control processor in the Vulcan Air Defense System M163A2 and M167A2, self-propelled and towed vehicles

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b. Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Fire Control Processor (FCP)
Servo Controller Board (SCB)
Test Program Set (TPS)
Vulcan Air Defense System (VADS)

6.6 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

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